

A PROJECT TITLE

ON

Player Performance Prediction Using Ridge and Lasso Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

Name: R . Radhakrishna **Roll:** 2520090063

Name: R. Jagadeesh **Roll:** 2520090030

Under the Esteemed guidance of

<Guide Name>

<Designation, Department>

CSIT



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Name of Title:

Player Performance Prediction Using Ridge and Lasso Regression

1. Abstract:

Player performance prediction has become an essential application of data analytics and machine learning in modern sports. Accurate prediction of player performance helps coaches, team managers, analysts, and selectors make informed decisions regarding team selection, player training, and match strategies. This project, titled "Player Performance Prediction Using Ridge and Lasso Regression," aims to develop a predictive model capable of estimating a player's future performance based on historical statistics and performance-related attributes.

The project utilizes a dataset containing various player performance indicators such as matches played, runs scored, batting average, strike rate, wickets taken, economy rate, catches, and other relevant features. Before model development, the dataset undergoes preprocessing, including handling missing values, removing duplicate records, feature scaling, and selecting significant attributes. Exploratory Data Analysis (EDA) is performed to understand the relationships among variables and identify trends that influence player performance.

Two regularized linear regression techniques, namely Ridge Regression and Lasso Regression, are implemented to build predictive models. Ridge Regression applies L2 regularization to reduce overfitting by shrinking coefficient values, while Lasso Regression uses L1 regularization to perform both coefficient shrinkage and automatic feature selection by eliminating less important variables. These methods improve model stability and prediction accuracy, particularly when dealing with multicollinearity and high-dimensional datasets.

The developed models are evaluated using standard performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) score. A comparative analysis of Ridge and Lasso Regression is conducted to determine the most effective approach for predicting player performance. The project also highlights the impact of feature selection and regularization on model accuracy and generalization.

Overall, this project demonstrates how machine learning techniques can be effectively applied to sports analytics for performance prediction. The proposed system provides a reliable, data-driven solution that assists sports organizations in evaluating player potential and making strategic decisions. Furthermore, the project establishes a foundation for future enhancements, including the integration of advanced machine learning algorithms, larger datasets, real-time performance tracking, and deep learning models to further improve prediction accuracy and decision-making in sports analytics.

2. INTRODUCTION:

Machine learning has become an important technology in sports analytics by helping predict player performance based on historical data. Accurate performance prediction enables coaches, selectors, and team management to make better decisions regarding team selection, player development, and match strategies. Instead of relying only on manual observation, machine learning models analyze large amounts of player statistics to identify patterns and estimate future performance.

The project "Player Performance Prediction Using Ridge and Lasso Regression" aims to predict a player's performance using historical performance data. The dataset contains various player-related features such as matches played, runs scored, batting average, strike rate, wickets taken, economy rate, catches, and other relevant statistics. These features are preprocessed and analyzed before training the prediction models.

This project uses two regression algorithms: Ridge Regression and Lasso Regression. Ridge Regression reduces overfitting by applying L2 regularization, while Lasso Regression uses L1 regularization to perform both regression and feature selection. These algorithms improve the model's prediction accuracy and help identify the most important factors affecting player performance.

The dataset is cleaned by handling missing values, removing duplicate records, and scaling the features where necessary. The trained models are evaluated using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-Squared (R^2) score. By comparing the performance of Ridge and Lasso Regression, the project identifies the model that provides better prediction accuracy.

This project demonstrates how machine learning techniques can be effectively applied in sports analytics to support data-driven decision-making and improve player performance evaluation.

3. SOFTWARE REQUIREMENTS:

Operating System

- Windows 10 / Windows 11

Programming Language

- Python 3.x

Development Environment

- Jupyter Notebook / Visual Studio Code (VS Code)

Python Libraries Used

- NumPy – Numerical computations
- Pandas – Data loading and preprocessing
- Matplotlib – Data visualization
- Seaborn – Statistical data visualization
- Scikit-learn (sklearn) – Machine learning algorithms (Ridge & Lasso Regression), model training, and evaluation

Dataset

- Player Performance Dataset (.csv)

Performance Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-Squared (R^2) Score

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: 500 MB free space

- Internet: Required for downloading libraries and datasets (optional after setup)

Signature of the Guide

Course Instructor